LLM orchestration

The missing layer b/w an LLM demo and a production AI app. An LLM generates text- an orchestrator builds intelligent systems.

① LLM orchestration
The layer that turns alone model into a real system- coordinating LLMs, tools, memory and rules to finish a job.

② Why it matters
A single prompt cracks the moment real users show up.
- orchestration adds control and repeatability
- Demos impress people - orchestration ships product

③ Prompt management
Prompts are built at runtime, not hardcoded strings.
- They adept to user role, task and history.
- Retrieved context is injected live

④ Tool calling
The system decides when to act instead of guessing.
- Search the web, query DBs, run code, call APIs
- Turns hallucination into verification

classmate
Date
Page

⑤ Retrieval- Augmented Generation
fetch trusted data before the model ever speaks.
- Grounded in your real documents
- Kills hallucinations on facts it never saw.

⑥ Memory
It remembers you across sessions, not just one chat.
- Preferences, history and workflow state
- "Book my usual" finally works.

⑦ Routing
Every request goes to the workflow built for it.
- Billing → Billing Agent
- Docs → RAG
- One door, many specialized rooms

⑧ State management
Know exactly where every multi-step task stands.
- Resume after a failure, skip nothing
- A checkout flow that survives a crash

⑨ Error handling
Real system fail- so u design for it
- Retries, timeouts, validation, fallbacks
- Fail safe and visible, never silent

(10) Prompt eng. vs Orchestration
One writes the script. The other directs the film.
- Prompt engineering → better instructions
- Orchestration → runs the whole system

(11) Sequential chains
each step hands its o/p to the next.
classify → retrieve → generate → validate

(12) Routing pattern
A router reads intent, then picks the best path.
- Right model, agent or workflow per task
- easy → cheap model
- hard → frontier model

(13) Parallel execution
Run independent tasks all at once.
- summarize · extract risks · find actions
- Same wait time, three times the work

(14) ReAct Pattern
The model reasons b/w every tool call.

Think → Act → Observe → Repeat → Answer

(15) Supervisor- Agent Architecture
A lead agent delegates to specialists.

Planner → Researcher → Writer → Criti...

(16) Human-in-the-loop
Pause for a human on the risky stuff.
- Refunds, outbound emails, payments
- "Approve $5k refund?" waits for you

(17) Customer support Assistant
One ticket, resolved end to end.

Classify → get A/C → search docs → draft → escalate

(18) Research Assistant
breaks a big question into answerable parts.

sub-questions → research → compare → cite

(19) Coding & sales copilots
Same orchestration, different domain.
- Code: read → fix → test → refine
- Sales: CRM → summarize → draft → next step

(20) The Payoff
This is what moves you from demo to product.
- Higher accuracy and reliability
- Scales up with a consistent experience

(21) The Hard Parts
Power has a price - go in with eyes open.
- more latency, cost and debugging
- state, safety and guardrails to manage

(22) Best Practices
Keep it boring and it stays reliable
- start simple; add steps only when needed
- validate, add fallbacks, trace every step